

东南大学考试卷 (A卷)

课程名称 操作系统原理及应用 考试学期 20-21-3 得分
适用专业 计算机类 考试形式 闭卷 考试时间长度 120 分钟

题号	一	二	三	四	五	六	七	八	总分
得分									
批阅人									

Answer the questions in the spaces provided on the question sheets.
If you run out of room for an answer, continue on the first page.

1. (20 points) Please answer the following True/False questions by marking "T" or "F".
- (a) With linked allocation, each file is a linked list of disk blocks.
(a) _____
 - (b) The optimal page replacement algorithm is the best choice in practice.
(b) _____
 - (c) Round Robin is a fair scheduling algorithm in that it gives each process an equal opportunity at the CPU.
(c) _____
 - (d) The operating system is responsible for resource allocation between competing processes.
(d) _____
 - (e) The long-term scheduler controls the degree of multi-programming, so it is invoked frequently.
(e) _____
 - (f) The interrupt vector contains the addresses of interrupted instruction.
(f) _____
 - (g) The disadvantage of spin locks include busy waiting and wasting CPU cycles when multiple processes share the same CPU.
(g) _____
 - (h) Threads are cheaper to create than processes.
(h) _____
 - (i) Binding of instructions and data to memory addresses can happen only at execution time.
(i) _____
 - (j) LRU and FIFO are cases of stack algorithm that do not have Belady's Anomaly.
(j) _____



(k) Polling, DMA and interrupt are three main interaction models between I/O controllers and CPUs to accomplish a complete I/O transfer.

(k) _____

(l) An individual threads will be discarded once it has completed its work, but the thread in the thread pool does not. It returns to the pool and awaits more work.

(l) _____

(m) Fixed-sized contiguous memory allocation suffers from internal fragmentation.

(m) _____

(n) Bit vector can be used for free disk space management in modern file systems.

(n) _____

(o) Shortest job first and priority based scheduling both could result in starvation.

(o) _____

(p) In One-to-One multi-threading model, a blocking kernel-scheduled thread blocks all threads in the same process.

(p) _____

(q) Suppose a two-level page table, on a translation look-aside buffer (TLB) miss, two loads from memory will be required to get the right translation information from the page table.

(q) _____

(r) The directory can be viewed as a symbol table that translates a file name into its directory entry which contains its low-level name (i.e., inode number).

(r) _____

(s) It's possible for all processes to complete without deadlock occurring when the system in an unsafe state.

(s) _____

(t) When using conditional variables and semaphores, programmers must be careful to avoid the program calls *signal()* before the corresponding wait().

(t) _____

2. (30 points) Short Answer Questions

(a) Compare the concepts of file, directory, and inode (or file control block).

(b) Describe the four necessary conditions of a deadlock.

(c) Compare the concepts of multi-programming and time sharing.



(d) List the timings when CPU scheduling takes place.

(e) List at least 5 issues considered in deciding the page size.

- (f) Suppose that a disk drive has 200 cylinders, numbered 0 to 199. The drive is currently serving a request at cylinder 62, and the previous request was at cylinder 73. The queue of pending requests, in FIFO order, is 13, 28, 87, 29, 129, 132, 183, 72, 46. Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests for each of the following disk-scheduling algorithms respectively?
- SSTF
 - SCAN
 - LOOK

3. (10 points) Scheduling

Consider the following set of processes, assumed to have arrived at time 0, in the order P1, P2, P3, P4, P5, with the length of the CPU-burst time given in milliseconds:

Process	P1	P2	P3	P4	P5
Burst Time	9	3	2	1	4
Priority	3	1	5	4	2

- (a) (6 points) Complete the following four Gantt charts that illustrate the execution of these processes using FCFS, SJF, non-preemptive priority (a smaller priority number implies a higher priority), and RR (quantum = 1) scheduling.

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19

P1

→ FCFS

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19

→ SJF

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19

→ n-p Priority

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19

P1:

P2:

P3:

P4:

P5:

→ RR

- (b) (4 points) What are the average turnaround time and the average waiting time of all process for each of the scheduling algorithms in part (a) respectively?

	FCFS	SJF	n-p Priority	RR
Average Turnaround Time				
Average Waiting Time				

4. (10 points) Deadlock Avoidance

Consider the following system snapshot using the data structures in the Banker's algorithm, with resources A, B, C, and D, and processes P_0 to P_4 :

Processes	Max				Allocation				Need				Available			
	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D
P_0	6	0	1	2	4	0	0	1								
P_1	1	7	5	0	1	1	0	0								
P_2	2	3	5	6	1	2	5	4								
P_3	1	6	5	3	0	6	3	3								
P_4	1	6	5	6	0	2	1	2								
Total													3	2	1	1

(a) (2 points) How many resources of type A, B, C, and D are there?

(a) _____

(b) (4 points) What are the contents of the Need and Available matrices? Fill in the above table.

(c) (2 points) Is the system in a safe state? Provide your reasons.

(d) (2 points) If a request from process P_4 arrives for additional resources of (1, 2, 0, 0), can the Banker's algorithm grant the request immediately? Show the new system state and other criteria.

5. (20 points) Memory Management

Consider the reference page sequence is 1, 2, 3, 4, 2, 3, 1, 4, 5, 6, 7, 4, 5, 6, 7, and the number of page frame is 3.

(a) (4 points) How many page faults for FIFO algorithm? Fill your answer in the blank, and complete the following diagram to show how the page faults are generated.

(a) _____

Ref String	1	2	3	4	2	3	1	4	5	6	7	4	5	6	7
Frame #1	1	1	1												
Frame #2	/	2	2												
Frame #3	/	/	3												
Page Faults	o	o	o												

(b) (4 points) How many page faults for LRU algorithm? Fill your answer in the blank, and complete the following diagram to show how the page faults are generated.

(b) _____

Ref String	1	2	3	4	2	3	1	4	5	6	7	4	5	6	7
Frame #1	1	1	1												
Frame #2	/	2	2												
Frame #3	/	/	3												
Page Faults	o	o	o												

(c) (4 points) How many page faults for OPT algorithm? Fill your answer in the blank, and complete the following diagram to show how the page faults are generated.

(c) _____

Ref String	1	2	3	4	2	3	1	4	5	6	7	4	5	6	7
Frame #1	1	1	1												
Frame #2	/	2	2												
Frame #3	/	/	3												
Page Faults	o	o	o												

(d) (2 points) Which of the three algorithm achieves the worst page fault rate?

(d) _____

(e) (2 points) Use this reference string to describe what is thrashing.

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(f) (4 points) With the reference string, how many frames are required at the timing of each reference. Use the working set model (with window size $\Delta = 5$) to provide your reason.

6. (10 points) Process Synchronization

The “H₂O” problem:

You have just been hired by Mother Nature to help her out with the chemical reaction to form water, which she does not seem to be able to get right due to synchronization problems. The trick is to get two *H* atoms and one *O* atom all together at the same time. The atoms are threads. Each *H* atom invokes a procedure `hReady` when it is ready to react, and each *O* atom invokes a procedure `oReady` when it is ready. For this problem, you are to write the code for `hReady` and `oReady`. The procedures must delay until there are at least two *H* atoms and one *O* atom present, and then one of the threads must call the procedure `makeWater` (which just prints out a debug message that water was made). After the `makeWater` call, two instances of `hReady` and one instance of `oReady` should return. Your solution should avoid starvation and busy-waiting.

To simplify this problem, you may assume that the semaphore implementation enforces FIFO order for wakeups — the thread waiting longest in `wait()` is always the next thread woken up by a call to `signal()`.

Provide the pseudo implementation of `hReady` and `oReady` using semaphores.

